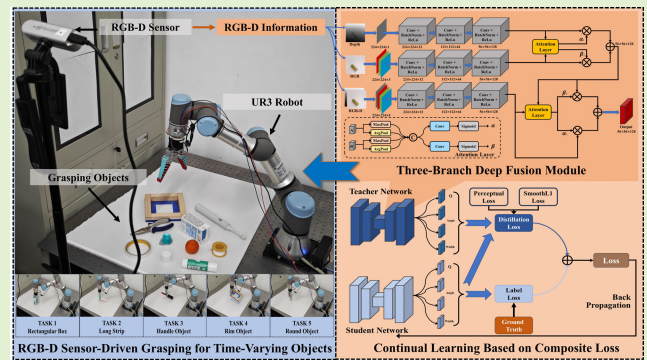


Robotic Grasping Detection Based on Continual Learning Using Perceptual Loss and Multibranch Deep Fusion

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Abstract—Vision-based grasping detection is extensively utilized in the field of production and manufacturing, leveraging multisource visual data to generate feature maps and achieve robust autonomous grasps. However, significant challenges remain in effectively integrating multisource visual inputs and overcoming catastrophic forgetting in scenarios that vary with time. To address these issues, this article proposes: 1) a three-branch RGB-D fusion module for cross-modal feature synthesis, integrated into the GR-ConvNet framework to optimize antipodal grasping detection; 2) a composite distillation strategy combining perceptual loss with smooth $L1$ loss to stabilize knowledge retention across sequential tasks; and 3) a robotic grasping detection system driven by RGB-D sensor integration to facilitate autonomous grasping of objects with diverse shapes. Comprehensive evaluations demonstrate state-of-the-art performance of our methods: 98.9% grasping detection accuracy on the Cornell dataset, 89.12% mean grasp accuracy on the final continual learning task, and 82% grasp success rate in real-world robotic trials. Moreover, ablation experiments conducted on our proposed model and the corresponding continual learning approach demonstrate the effectiveness of the three-branch deep fusion (3-BDF) module and the combined distillation loss. To our knowledge, this is the first application of a perceptual loss approach in RGB-D sensor-driven grasping detection tasks designed for continuously changing scenarios. Code and Video are available at: <https://github.com/lyxhnu/Cornell-CL>

Index Terms—Continual learning, grasp detection, RGB-D fusion, RGB-D sensors, robotic grasp.



I. INTRODUCTION

GRASP detection, as a general vision sensors-based measurement technology, is widely used in the field of production and manufacturing. This technique typically utilizes vision-based perception to obtain the position and shape information of the target, and then applies sensor analysis to generate the appropriate grasp pose, enabling the robot to perform the grasping action [1]. After obtaining the grasp pose, the robot moves its end-effector to the target grasp position through path planning and coordinate transformation,

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and then executes the grasping action using the end effector's gripper [2], [3]. In early research on grasping detection, object detection networks were primarily relied upon to generate grasp rectangles [4], [5]. This data-driven discriminative technique enables rapid grasp detection in specific scenarios. However, in recent years, this indirect approach to grasp planning has gradually been replaced by generative grasping detection methods, as they can directly generate feature maps from visual information to guide robotic grasping [4], [6]. With the continuous development of these generative grasp detection methods, using a single dataset to implement robotic grasp detection in a controlled real-world scenario is no longer challenging. Besides, these methods are still constrained by several limitations, such as heavy dependence on datasets, inefficiency in processing multisource visual data, and difficulties in adapting to dynamically changing grasping detection scenarios [7], [8], [9]. Therefore, fully leveraging datasets from multiple sources and adapting grasp perception to different scenarios have become the primary challenges in developing grasp detection methods. Recent studies on robotic

grasping detection primarily analyzed multisource information, particularly RGB-D data, which includes both color images and depth information. Depth data is typically captured using infrared sensors based on the binocular vision principle, while color data is directly obtained through RGB sensors. These complementary acquisition approaches compensate for spatial details overlooked by each respective method, making RGB-D data superior to conventional visual imagery in terms of visual content richness. Therefore, this type of data not only improves grasp success rate but also aids in planning the motion path for robots. Maximizing the utility of RGB-D information within the smallest feasible model scale enhances both the accuracy and speed of grasp detection. This enables deployment of grasp detection methods across a broader range of scenarios. However, effectively leveraging RGB-D data remains a challenge in the field of grasp detection. Current generative grasping detection methods mainly rely on simple stacking of RGB and depth images to combine information from both sources [10], [11], [12]. In addition, some methods explore early or late fusion from separate branches for RGB and depth information [1], [12], [13], [14]. However, these approaches have not considered intermediate fusion across multiple branches. Moreover, some fusion methods are too complex, making it difficult to integrate them with other modular methods.

On the other hand, current generative grasping detection models are highly dependent on specific grasp detection datasets, with each model trained on a single dataset. However, in real-world environments, the objects to be grasped or the grasp scenarios continuously change over time. Thus, the data-driven detection method must constantly adapt to time-varying visual information. When a grasping detection system is required to adapt to multiple sequential datasets, catastrophic forgetting often occurs, causing them to lose knowledge acquired from earlier training [15], [16], [17].

To overcome the challenge of analyzing multisource information, this article proposes a three-branch deep fusion (3-BDF) module. This module fully exploits multisource features by progressively fusing RGB, depth, and RGB-D information during the feature extraction phase of the detection model. Furthermore, to enable the grasping detection system to continuously adapt to different types of visual information over time, we propose an advanced continual learning method based on a combined distillation loss. Compared to recent continual learning methods, this method better solves the problem of catastrophic forgetting. Finally, we apply the continually learned model to a collaborative robotic system based on visual servo grasping detection. By comparing the grasp success rates of different methods across multiple real-world tasks, our method achieves optimal performance in this detection system. In summary, the contributions of this article are as follows.

- 1) This work proposes a 3-BDF module used on the GR-ConvNet [18], designed to effectively exploit the visual information captured by RGB-D sensors. Compared to its original version, the model achieves a more than 1.2% improvement in grasping detection accuracy on the Cornell dataset, reaching an accuracy of 98.9% with

a computational cost of 26.4 GFlops and the number of parameters of 2.39M. Compared with other advanced grasping detection models, our method can achieve the most competitive accuracy.

- 2) We propose a continual learning model incorporating the combined distillation loss to address the challenges faced by existing grasp detection methods in adapting to time-varying data. By distilling both high-level semantic information and output features of the model, our method achieves an average accuracy of 89.12% in the final continual learning task. Compared to other continual learning methods, the proposed model attains the highest average accuracy while demonstrating a forgetting rate comparable to that of replay-based methods.
- 3) We conducted extensive testing on a real robotic grasping detection system based on the UR3 robot. In scenarios where both the grasping scenes and objects were previously unseen, we achieved an average grasp success rate of 82% after testing across five tasks. Moreover, this method significantly outperforms other comparable methods in terms of grasp success rate.

II. RELATED WORKS

A. RGB-D Information Fusion Methods for Robotic Grasping Detection

In existing grasping detection research, the use of RGB-D data has become mainstream. This type of data is typically acquired through RGB-D sensors and integrates both color and depth visual information, which is commonly represented in image-based formats. Most grasping detection methods primarily focus on directly stacking RGB and depth images to form an RGB-D input for grasp detection networks [2], [9]. However, this straightforward method does not take into account the correspondence between the two types of information. Currently, the fusion of RGB and depth information can be categorized into early fusion, intermediate fusion, and late fusion. Early fusion integrates the two types of information before the model's feature extraction, with existing models that stack RGB and depth images falling into this category. Late fusion, on the other hand, occurs at the decision layer of the model. Intermediate fusion combines the characteristics of both, integrating the information during the intermediate stage of the model's inference. Recent researches in robotic grasping detection that use RGB-D information fusion mostly fall under the intermediate fusion category. The pixel-wise RGB-D dense fusion method proposed by Tian et al. [1] effectively explores the feature relationships between RGB and depth information, performing dense fusion at the intermediate layers. However, this method requires significant computational resources to support the fusion process. In addition, Song et al. [13] propose a hierarchical fusion method for RGB and depth information, which also incorporates late fusion of RGB-D information. However, this approach requires a fully convolutional network for computation at each branch, leading to high computational demands. Furthermore, Pei et al. [12] propose a multimodal fusion method using a channel and spatial (MCS) attention module, which relies on channel-switching techniques for multisource information fusion. However, this

method requires the reuse of the MCS module to achieve significant performance improvement. On the other hand, RGB-D fusion techniques have also been widely adopted in the field of salient object detection. The deep learning architectures commonly employed in this area share similarities with those used in grasp detection. However, these fusion methods often rely on pretrained backbone networks such as ResNet50 or Swin Transformer for feature extraction, which require substantial computational resources and struggle to meet the real-time demands of robotic grasping [19]. On the contrary, using lightweight pretrained models tends to result in insufficient accuracy [20]. Therefore, designing a simple and easily transferable multisource information fusion module remains a challenge in the field of robotic grasping detection based on RGB-D information.

B. Continual Learning Methods for Robotic Grasping Detection

Within the realm of grasping detection, a robot's visual system is continually required to adapt to new grasp targets. How to enable visual sensors to continuously adapt to evolving grasp targets over time remains a significant challenge. Continual learning, also referred to as lifelong learning and incremental learning, as an online learning methodology, is used to enhance the adaptability of visual measurement systems to evolving data streams [17], [21]. In robotic grasping detection, continual learning methods based on static architectures are typically employed since they do not need to address class-incremental tasks [22]. For continual learning methods under this framework, they can generally be categorized into regularization-based methods, replay-based methods, and distillation-based methods [15]. Regularization-based methods typically mitigate catastrophic forgetting by penalizing the loss function to constrain model parameter updates, as exemplified by the synaptic intelligence (SI) method proposed by Zenke et al. [23]. However, such methods exhibit limited applicability and may fail to achieve adequate learning capacity for new tasks in certain scenarios [22]. Furthermore, replay-based methods directly expose the model to past training data during the current task, thereby enabling the consolidation of prior knowledge. While these methods can significantly enhance model accuracy and substantially mitigate forgetting, they often impose substantial storage overhead [22], [24]. This is not applicable to a commonly used visual measurement system. Moreover, the selection of samples to retain remains a critical challenge, as suboptimal sample selection can severely degrade the algorithm's training outcomes [7]. In contrast, distillation-based methods neither incur significant memory overhead nor risk data leakage from past tasks. As a model-centric strategy, these methods mitigate catastrophic forgetting during adaptation to new tasks by distilling critical experience from trained models. By establishing the continual learning frameworks for robotic grasping detection, Yang et al. [9] and Auddy et al. [24] preliminarily demonstrate that replay-based methods outperform regularization-based methods in solving this problem. Besides, Liu et al. [7] introduced a continual learning framework featuring a knowledge-selective module. While this method achieves superior grasping detection

performance compared to replay-based methods via an innovative distillation mechanism, its effectiveness hinges on extensive hyper-parameter optimization.

III. PRELIMINARY

A. Statement of Grasping

This article investigates 2-D planar robotic grasping estimation utilizing RGB-D information captured by visual servo systems. To better facilitate grasp planning for a robot utilizing visual measurement, building upon the most widely adopted 5-D grasp pose representation [25], a grasp configuration can be parameterized as

$$\mathbf{g}_r = (x_r, y_r, q_r, \theta_r, \omega_r) \quad (1)$$

where (x_r, y_r) represents the coordinate of the center position of a grasp pose, q_r is the grasp quality score which can quantify the likelihood of successful robotic grasp, θ_r demonstrates the rotation angle of the end-effector when the robot grasping, and ω_r indicates the width at which the gripper opens. Specifically, the grasp quality score is confined to the interval $[0, 1]$, with values approaching 1 indicating more favorable grasps. The original framework [25] prescribes grasp angles within $[-\pi/2, \pi/2]$ radians and grasp widths spanning 0 to 150 units. Notably, the upper width limit (150 units) can be adapted to match the maximum aperture of individual grippers.

B. Definition of Continual Learning Tasks for Robotic Grasping Detection

Continual learning grasp detection tasks differ from ordinary grasp detection tasks. Such tasks require continuously handling new tasks arriving over time. According to Liu et al. [7], this process can be defined as

$$T_\lambda = f(T_{\lambda-1}, D_\lambda), \lambda \in \{2, 3, \dots, \mu\} \quad (2)$$

where $f(\cdot)$ represents a continual learning method for training a model T in the task λ , and D_λ means the trainable data belonging to the task λ . This formulation captures how the continual learning framework updates model T_λ during task λ training by using the teacher model (the student model trained in task $\lambda-1$) with the current task's accessible training dataset D_λ .

IV. METHODS

A. 3-BDF Module: A Three-Branch Structure for Fusing RGB-D Information Deeply

Currently, grasping detection networks predominantly implement RGB-D fusion through channel-wise concatenation. On the one hand, fusion methods employing channel-wise concatenation, whether through early or middle fusion, struggle to effectively capture spatial dependencies between different modalities. On the other hand, current approaches that facilitate thorough interaction along the channel dimensions of both modalities often incur high computational costs. Whether through channel exchange [12], channel concatenation, or transformer-based architectures [26], these methods

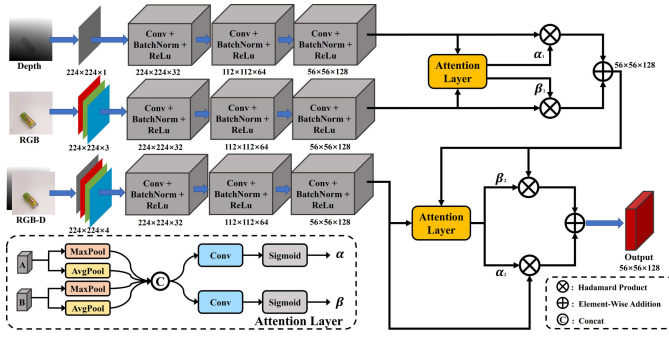


Fig. 1. Structure of 3-BDF Module.

substantially increase the computational burden on the network.

To mitigate unnecessary computational costs arising from excessive channel-wise interactions, we introduce a third modality input derived from the network's original inputs. This input is constructed by simply concatenating the RGB and depth images along the channel dimension. Moreover, a triple-branch structure is adopted to effectively capture spatial dependencies among different input modalities. Each branch maintains identical network components while preserving operational independence. The outputs are then fused through an attention-guided weighting mechanism operating across spatial dimensions. Fig. 1 schematically details the architectural configuration.

The effectiveness of our proposed structure primarily stems from the attention mechanism's capability to filter and prioritize effective information. In our attention mechanism, two feature maps are utilized to generate weights through pooling and stacking operations. Specifically, each input feature map undergoes average pooling and max pooling separately, resulting in four outputs. These outputs are then concatenated along the channel dimension. Subsequently, each branch passes through a convolutional layer followed by a sigmoid activation function to produce the corresponding weights, α and β . Subsequently, these weights are used to perform a weighted summation on the corresponding features. This approach leverages the attention module to effectively suppress less important information from both the color and depth images, allowing the fused multimodal information to focus more on key features and improving the overall accuracy and effectiveness of information processing. These processes can be expressed by the following equations:

$$\beta, \alpha_1 = \text{attention}(R, D) \quad (3)$$

where R represents the RGB image information and D represents the depth image information. Once the weights are obtained, the following equation can be used to fuse the feature information from the depth and color images:

$$F = \beta_1 \times R + \alpha_1 \times D. \quad (4)$$

However, merging the two channels alone still does not fully leverage the feature information. Therefore, we employ an attention module to combine the fused feature map with

the features extracted from a third branch

$$\begin{aligned} \beta_2, \alpha_2 &= \text{attention}(RD, F) \\ \text{output} &= \beta_2 \times RD + \alpha_2 \times F \end{aligned} \quad (5)$$

where RD denotes the output of the feature extraction module after stacking the RGB and depth images, and the output represents the final output of the 3-BDF module.

B. Network Architecture

To construct a high-performance grasp detection model, we adopt the widely used GR-ConvNet architecture [6] as the foundation. We replace its front-end feature extraction structure with the 3-BDF module while retaining subsequent layers consistent with GR-ConvNet. This architecture serves to facilitate the comparative assessment of our proposed structure's advantages while efficiently harnessing the generative mechanism integrated into the grasp detection framework. The features processed through the 3-BDF module are fed into residual blocks, followed by transposed convolution upsampling to generate output representations: grasp quality map, gripper width map, grasp angle sine map, and grasp angle cosine map for synthesizing grasp configurations. Owing to the performance improvement of this model primarily hinges upon the 3-BDF module, we designate it as DF-ConvNet, where DF denotes deep fusion. The overall framework of the designed grasp detection model (DF-ConvNet) is presented in Fig. 2.

C. Continual Learning Framework and Composite Distillation Loss

To address the dynamic changes in object morphological characteristics during grasping detection, we adopt a continual learning architecture inspired by Liu et al. [7] to design our training strategy, as illustrated in Fig. 3. This strategy is built upon a teacher-student framework, ensuring that both the teacher and student models share identical structures. By leveraging the extensive knowledge and mature expertise embedded in the teacher model, the student model's learning process is precisely guided and effectively regulated. This approach significantly mitigates the student model's tendency to forget previously acquired knowledge during the progression of continual learning. Besides, when initiating training for a new task under this continual learning strategy, the best-performing model from the previous task is selected to serve as both the teacher model for the subsequent task and the initialization base for the student model. Moreover, throughout the training process, the teacher model remains frozen with its internal parameters not undergoing further updates.

In existing continual learning frameworks, the loss function combines distillation loss with label loss. The distillation loss transfers the teacher model's task-specific knowledge, while the label loss quantifies the discrepancy between the student model's predictions and ground-truth annotations. However, in existing methods, distillation loss often requires tuning multiple hyperparameters to combine with label loss, thereby preventing the model from overfitting to the outputs of the teacher model. On the other hand, many existing knowledge distillation models were originally developed for image

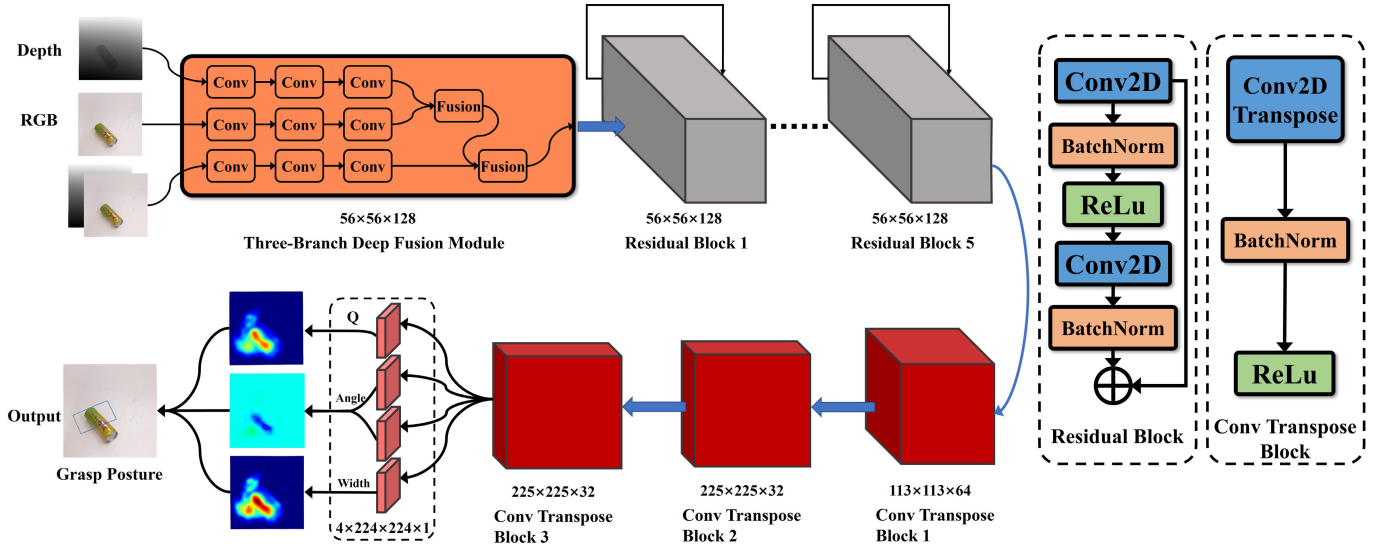


Fig. 2. Overall of the DF-ConvNet architecture. The model employs a 3-BDF module to extract and fuse feature information, leveraging the residual layers and deconvolutional layers of GR-ConvNet to generate grasp poses.

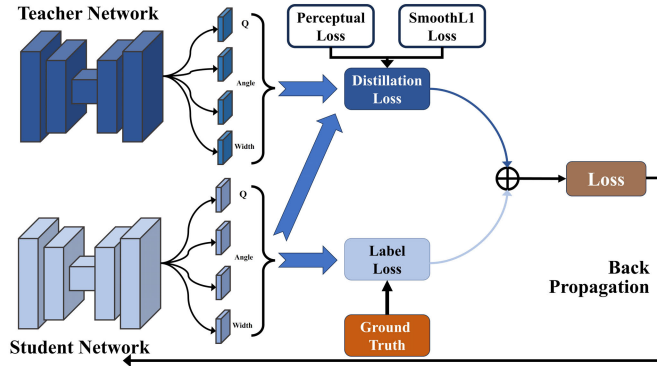


Fig. 3. Training workflow in continual learning. The algorithm enables a grasping detection system to dynamically adjust to visual measurement tasks that evolve over time.

classification tasks. These models typically rely on straight-forward KL divergence and cross-entropy loss to measure distributional differences, rather than absolute discrepancies, in label predictions. However, due to architectural differences, such methods cannot be directly applied to generative grasp detection models to assess distribution disparities between labels and predictions. This limitation may lead to conflicts between old and new knowledge during learning, ultimately undermining the model's ability to balance plasticity and stability in continual learning scenarios.

To address these issues, we propose to integrate a perceptual loss function [27], [28], derived from the field of style transfer, that extracts feature representations of output images from a pretrained model to quantify the differences between model outputs. This loss function captures high-level semantic features, allowing it to measure output discrepancies between the teacher and student models at a holistic level. Moreover, based on this characteristic, this loss function is also particularly suited for addressing the challenge of excessive hyperparameter tuning required by distillation loss. We propose to compute the distillation loss through a combination

of perceptual loss and smooth $L1$ loss, leveraging features extracted by a pretrained convolutional neural network from the grasping network's outputs. This method quantifies the discrepancy between teacher and student models by evaluating feature-level differences in their predictions. However, quantifying the feature differences between the outputs of the student network and the teacher network on a pretrained network poses a challenge. Given that the smooth $L1$ loss efficiently computes pixel-level differences between two feature matrices, we propose employing the smooth $L1$ loss to measure the discrepancies in the features extracted by the pretrained network. Specifically, the smooth $L1$ loss is formulated as follows:

$$L1 = \begin{cases} 0.5(\mathbf{F} - \mathbf{F}')^2, & \text{if } |\mathbf{F} - \mathbf{F}'| < 1 \\ |\mathbf{F} - \mathbf{F}'| - 0.5, & \text{otherwise} \end{cases} \quad (6)$$

where $\mathbf{F}$ indicates the feature matrix output by a model. Since each grasp feature map generated by the grasp network contains only one channel dimension, we propose two adaptation strategies for pretrained models. The first method modifies the output channels of the initial convolutional layer in the pretrained network while retaining all subsequent layers unchanged. Moreover, the second strategy replicates the input feature maps and concatenates them along the channel dimension to emulate standard RGB image inputs. To identify the optimal adaptation strategy, we conducted a series of ablation studies prior to comparative experiments. The results demonstrated that the first strategy achieved better performance than the other. In terms of network architecture, VGG19 outperformed VGG16, while VGG16 surpassed MobileNetV2 in effectiveness. Moreover, all pretrained models employed in our study are derived from the official PyTorch versions [29] pretrained on the ImageNet-1k dataset [30]. Furthermore, experiments revealed that increasing the number of layers for perceptual loss computation did not proportionally enhance performance. Consequently, we selected the 19th layer of VGG19 for perceptual loss calculation.

It should be noted that our proposed composite loss includes both perceptual loss quantified by smooth $L1$, and direct pixel-wise smooth $L1$ comparisons between student and teacher outputs. Empirical evaluations demonstrate that mean squared error (mse) optimizes performance when perceptual loss alone is applied as the distillation loss, while smooth $L1$ -based perceptual loss achieves superior results in the composite loss configuration. Specifically, a perceptual loss function based on the smooth $L1$ loss can be formulated as follows:

$$L_{P_i}(\mathbf{O}_S, \mathbf{O}_T) = \frac{\sum_{j=1}^{C_i} \sum_{h=1}^{H_i} \sum_{w=1}^{W_i} L1(\varphi_{i,j,h,w}(\mathbf{O}_S) - \varphi_{i,j,h,w}(\mathbf{O}_T))}{C_i H_i W_i} \quad (7)$$

where $\mathbf{O}_S$ represents the output of the student network, $\mathbf{O}_T$ denotes the output of the teacher network, $\varphi_i(\mathbf{O}_S)$ corresponds to the feature maps generated by processing $\mathbf{O}_S$ through the i th layer of the VGG19 model, and $\varphi_i(\mathbf{O}_T)$ similarly refers to the feature maps obtained when $\mathbf{O}_T$ passes through the same VGG19 layers. Moreover, $\varphi_{i,j,h,w}(\cdot)$ indicates the element at row h , column w of the j th channel in the feature map. Therefore, the distillation loss can be expressed as

$$L_{q/\theta/\omega}^{\text{distillation}} = L_{P_i}(\mathbf{O}_S, \mathbf{O}_T) + L1(\mathbf{O}_S, \mathbf{O}_T). \quad (8)$$

The label loss components, including grasp quality, sine/cosine values of grasp angles, and grasp width map, are all computed using the smooth $L1$ loss function. The complete formulation of the label loss can be expressed as follows:

$$L_{q/\theta/\omega}^{\text{label}} = L1(\mathbf{O}_S, \mathbf{O}_L) \quad (9)$$

where $\mathbf{O}_L$ represents the corresponding feature maps generated by the labels. Thus, the loss function in training can be ultimately expressed as

$$L_{q/\theta/\omega} = L_{q/\theta/\omega}^{\text{distillation}} + L_{q/\theta/\omega}^{\text{label}}. \quad (10)$$

D. Evaluation Criteria

Based on the commonly utilized evaluation standards [6], [25], the Jaccard index and the angle threshold are adopted by us to evaluate whether an object has been successfully grasped

$$\begin{cases} |\theta_I - \theta_A| < 30^\circ \\ \text{IoU}(\mathbf{g}_I, \mathbf{g}_A) = \frac{\mathbf{g}_I \cap \mathbf{g}_A}{\mathbf{g}_I \cup \mathbf{g}_A} > 0.25. \end{cases} \quad (11)$$

In (11), θ_I denotes the grasp angle predicted by the model, while θ_A represents the actual grasp angle from the corresponding ground truth. In addition, $\mathbf{g}_I$ refers to the grasp rectangle generated by the model, whereas $\mathbf{g}_A$ corresponds to the actual grasp rectangle in the ground truth. On the other hand, in the context of continual learning for robotic grasping detection, the mean grasp accuracy is commonly adopted as a metric to evaluate the performance of a grasping detection model [15], [22]. Therefore, the mean grasp accuracy $\bar{E}$ of the task $\lambda - 1$ can be represented as

$$\bar{E}_{\lambda-1} = \frac{1}{\lambda-1} \sum_{a=1}^{\lambda-1} E_{\lambda-1,a} \quad (12)$$

TABLE I
HARDWARE AND SOFTWARE

Configuration Item	Server for Training	System for Test
System	Ubuntu 20.04 LTS	Ubuntu 20.04 LTS
CPU	13th Gen Intel(R) Core(TM) i9-13900K	13th Gen Intel(R) Core(TM) i7-13650HX
GPU	NVIDIA GeForce RTX 4090 (24G)	NVIDIA GeForce RTX 4060 Laptop
RAM	128G	24G
CUDA	12.1	11.0
Cudnn	8.9.2	8.0.4
OpenCV	4.9.0	4.4.0
Pytorch	2.2.1	1.7.1
Torchvision	0.18.0	0.8.2
Python	3.11.5	3.6.13
Robot	-	UR3
Vision Sensor	-	RealSense D435

where $E_{\lambda-1,a}$ denotes the grasp success rate of the model tested in task a when a total of $\lambda - 1$ tasks have been trained by this model. In addition, backward transfer (BF) can be utilized to quantify the degree of forgetting in a model. The backward transfer after training on the a th task can be defined as follows:

$$\text{BF}_a^\lambda = \frac{1}{a} \sum_{i=1}^a (E_{i,i} - E_{a,i}), \quad i < a. \quad (13)$$

V. EXPERIMENTS

A. Experimental Settings and Software–Hardware Configuration

Our models' training processes in this article were executed on a server configuration featuring a 13th Gen Intel¹ Core² i9-13900K processor and NVIDIA GeForce RTX 4090 GPU. All architectures were developed and trained using the PyTorch framework, with subsequent robotic validation experiments implemented on an Ubuntu 20.04-based laptop system. Our robotic grasping detection system is loaded on a UR3 collaborative robot from Universal Robots, controlled through ROS Noetic's MoveIt package. The visual measurement is facilitated by a RealSense D435 camera module. Comprehensive specifications of both hardware components and software environments are systematically presented in Table I.

The proposed DF-ConvNet requires validation through three experimental phases: fundamental grasping detection verification, continual learning evaluation, and robotic system test. For fundamental grasping detection verification, we trained DF-ConvNet on the Cornell dataset using a 9:1 training-test split ratio with fivefold cross-validation. Training employed the Adam optimizer with hyperparameters set to learning rate = $1e^{-3}$, batch size = 8, and 50 epochs, aligning with common experimental configurations [10], [11], [12]. To further validate our methods, we conducted experiments on the large-scale grasping dataset Jacquard [31] using a 9:1 train-test split.

¹Registered trademark.

²Trademarked.

TABLE II
PERFORMANCE COMPARISON IN CORNELL DATASET

Method	Input Size	Input	Params	Acc(%)
(2020) GGCNN [25]	300 × 300	RGB-D	72K	73.9
(2020) GR-ConvNet [6]	224 × 224	RGB-D	1.9M	97.7
(2022) SE-ResUNet [32]	224 × 224	RGB-D	24.0M	98.2
(2022) TF-Grasp [26]	224 × 224	RGB-D	-	98.0
(2023) Efficient Grasp [33]	224 × 224	RGB-D	1.2M	97.8
(2024) BA-Grasp [14]	224 × 224	RGB-D	-	97.8
(2024) QQGNN1 [10]	300 × 300	RGB-D	238K	97.7
(2024) FANet-GPU [34]	224 × 224	RGB-D	-	98.5
(2024) MCS-ResNet [12]	224 × 224	RGB-D	7M	98.5
(2025) RAMPGrasp [35]	224 × 224	RGB-D	-	98.4
DF-ConvNet	224 × 224	RGB-D	2.4M	98.9

TABLE III
ABLATION STUDY IN CORNELL DATASET

Method	Params	Acc(%)
(1-Branch Module) GR-ConvNet [6]	1.9M	97.7
2-Branch Module	2.2M	98.2
(3-Branch Module) DF-ConvNet	2.4M	98.9

To accommodate the experimental conditions of this dataset, we adjusted the learning rate and the total number of training epochs to $1e^{-4}$ and 100, respectively. Given that Jacquard is a large-scale dataset, comprising over 11-k objects and 50-k images, cross-validation was performed exclusively on the Cornell dataset and not on Jacquard. It should be noted that, whereas the input image size was set to 224×224 for the Cornell dataset, a size of 300×300 was used for Jacquard in order to align with common experimental conventions.

In continual learning scenarios, hyper-parameter settings followed established conventions from [7], utilizing identical Adam optimizer parameters (learning rate = $1e^{-3}$, batch size = 8) with 15 epochs per task. The continual learning dataset derived from the Cornell dataset followed the partitioning methodology described in [7] and [24], maintaining 75:25 train-test ratios across sequential tasks.

B. Fundamental Grasping Detection Experiments

To validate the effectiveness of our 3-BDF module and assess DF-ConvNet's performance, comprehensive evaluations were conducted on the widely adopted Cornell dataset. Experimental configurations aligned with the specifications outlined in Section V-A, incorporating online augmentation techniques such as random rotation, scaling, and brightness variation for image preprocessing. To minimize the randomness of experimental results, we employed a fivefold cross-validation approach to obtain the experimental outcomes. Comparative outcomes across benchmark models are systematically summarized in Table II. Moreover, since most studies do not explicitly provide the computational complexity of their models, we have omitted this metric from the table (the computational cost of our model is 26.4 GFlops). From the results, it can be observed that our method achieves a 1.2% improvement in performance compared to the original GR-ConvNet, with only a 0.5M increase in the number of parameters. In comparison to other state-of-the-art models proposed in recent years, our model demonstrates highly competitive results in balancing parameter efficiency and grasping detection performance.

TABLE IV
PERFORMANCE COMPARISON IN JACQUARD DATASET

Method	Input Size	Input	Acc(%)
(2020) GGCNN [25]	300 × 300	RGB-D	78.0
(2020) GR-ConvNet [6]	224 × 224	RGB-D	94.6
(2022) TF-Grasp [26]	224 × 224	RGB-D	94.6
(2023) Efficient Grasp [33]	300 × 300	RGB-D	93.6
(2024) BA-Grasp [14]	-	RGB-D	94.6
(2025) RAMPGrasp [35]	320 × 320	RGB-D	95.6
DF-ConvNet	300 × 300	RGB-D	95.6

Besides, we conducted an ablation study on the Cornell dataset to validate the effectiveness of the 3-BDF module. Models with different numbers of branches were systematically compared. The single-branch model follows the original GR-ConvNet architecture, where RGB and depth information are concatenated at the channel level for feature extraction. The dual-branch variant utilizes two separate streams for RGB and depth inputs, integrating their features through the fusion strategy depicted in Fig. 1. The triple-branch structure corresponds to the proposed DF-ConvNet in this work. All experimental settings remained consistent with those reported in Table II, and the results of the ablation experiments are summarized in Table III. From this experiment, it can be observed that model performance consistently improves with the increase in the number of fusion branches, while each additional branch introduces only approximately 0.2 million parameters. By expanding from one to three branches, the grasping performance advances from the level of GR-ConvNet (proposed in 2020) to the state of the art.

To further compare the differences between our method and other models, we also conducted evaluations on the Jacquard dataset under the same configuration described in Section V-A. Since different methods vary significantly in terms of input size on this dataset, we adopted an input size of 300×300 to align with common practice. For a method whose input size was not explicitly reported in the literature, a "-" is indicated in the results. Table IV presents the comparative outcomes on this dataset. The compared methods are largely drawn from those already listed in Table II. Moreover, some approaches were excluded due to differences in dataset usage, input types, or data splitting methods. As the results indicate, larger scale networks generally achieve superior grasping performance on this dataset. Compared to larger scale models, our approach utilizes the 3-BDF module to effectively integrate complementary information from RGB and depth images, enabling focused attention on relevant regions. Therefore, our proposed DF-ConvNet attains optimal performance comparable to the recent transformer-based method (RAMPGrasp), demonstrating that our approach remains competitive comparable to larger models in large-scale datasets for grasp detection.

C. Experiments of Continual Learning Methods

In real-world scenarios, training a grasp model becomes increasingly challenging over time. While approaches such as data augmentation and expanding the raw data samples can help mitigate disturbances caused by variations in illumination, color, and viewpoint, they often remain insufficient in

TABLE V
CONTINUAL LEARNING PERFORMANCE COMPARISON OF MEAN
GRASPING DETECTION ACCURACY

Method	Mean Grasp Accuracy (%)				
	Task1	Task2	Task3	Task4	Task5
Liu et. al	97.13	92.14	93.02	82.16	67.00
GR + SI	96.05	90.58	92.22	83.59	64.64
GR + Memory Replay	96.05	86.67	88.29	79.49	81.55
GR + Smooth L1 Loss	96.05	96.05	92.36	89.71	75.36
GR + Composite Loss	96.05	95.85	94.85	90.40	84.89
Ours + SI	98.68	91.01	94.09	88.09	63.06
Ours + Memory Replay	98.68	91.01	91.04	81.99	82.24
Ours + Smooth L1 Loss	98.68	95.85	93.82	91.64	75.72
Ours + Perceptual Loss	98.68	96.93	95.57	91.06	88.89
Ours + Composite Loss	98.68	98.02	96.96	93.95	89.12

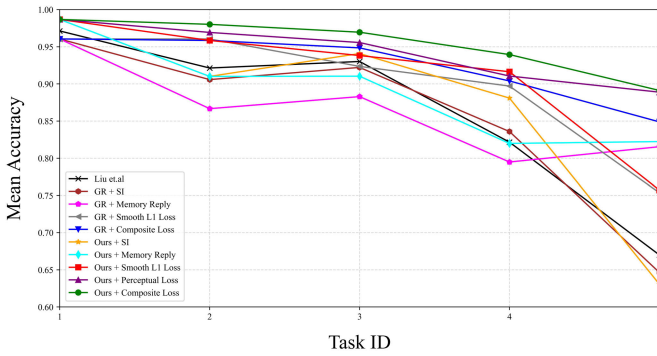


Fig. 4. Mean grasping detection accuracy results of different models paired with various continual learning methods across five tasks.

addressing changes in the appearance characteristics of objects to be grasped. A continual learning version of the Cornell dataset was introduced to address such challenges [24]. The classification of distinct tasks is primarily determined by the geometric characteristics of target objects, which are categorized into five specific groups: rectangular box, long strip, handle object, rim object, and round object. To objectively evaluate the performance of our method in continual learning tasks, we adopted the dataset configuration consistent with studies [7], [24]. During the training process, the sequence of tasks was maintained consistent with that described in the aforementioned studies. In addition, the experimental hyperparameters were set in alignment with those in study [7]. It is worth noting that the student network used in each task differs from that in study [7]. Specifically, it employs a student network initialized with fresh parameters for each task, whereas our approach utilizes parameters inherited from the student network trained in the previous task.

To control for a single variable in the evaluation, we paired DF-ConvNet with various continual learning methods and conducted experiments by combining these methods with the classic GR-ConvNet. Given the scarcity of comparable studies integrating grasp prediction models with continual learning frameworks in this dataset, our comparative analysis was primarily restricted to the methodological approach proposed by Liu et al. [7]. Furthermore, for performance evaluation of the continual learning architectures in these experiments, we primarily utilize the two metrics defined by (12) and (13).

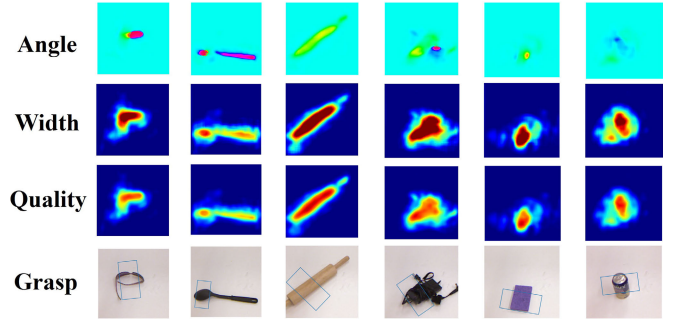


Fig. 5. Visually presented demonstration of DF-ConvNet's grasping detection with RGB-D data after being trained on five continual learning tasks.

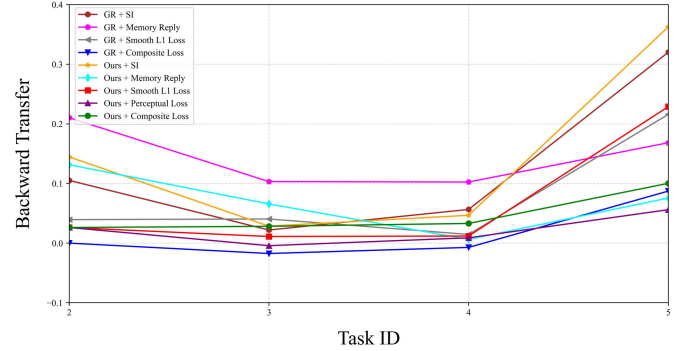


Fig. 6. Backward transfer (BF) results of different models paired with various continual learning methods across five tasks.

The grasp success rate is derived from (11) as specified in our computational framework. Table V summarizes the mean grasp accuracy results across different method configurations. Specifically, our grasp detection model primarily compares with GR-ConvNet, while continual learning approaches are benchmarked against SI-based methods [22], [23], memory replay-based strategies (20% replay) [15], [24], and knowledge distillation variants employing different loss functions. The experiments comparing our model with perceptual loss, smooth $L1$ loss, and composite loss configurations serve as ablation studies to validate the effectiveness of our composite loss design. Furthermore, to validate the effectiveness of our continual learning method, comparative experiments were conducted by integrating our model with various continual learning strategies. Within identical continual learning frameworks, comparisons between our model and GR-ConvNet aim to verify the enhanced performance of our RGB-D fusion architecture. In addition, comparisons between our proposed method and the method proposed by Liu et al. [7] were carried out to demonstrate the superiority of our combined approach.

Fig. 4 visually compares the performance metrics from Table V using the line graph to highlight distinctions among different approaches. Ablation studies on distillation loss variants reveal that our composite loss demonstrates superior performance compared to individual loss types, with perceptual loss alone outperforming smooth $L1$ loss in isolated configurations. Comparative experiments confirm that our continual learning method achieves significantly better results than

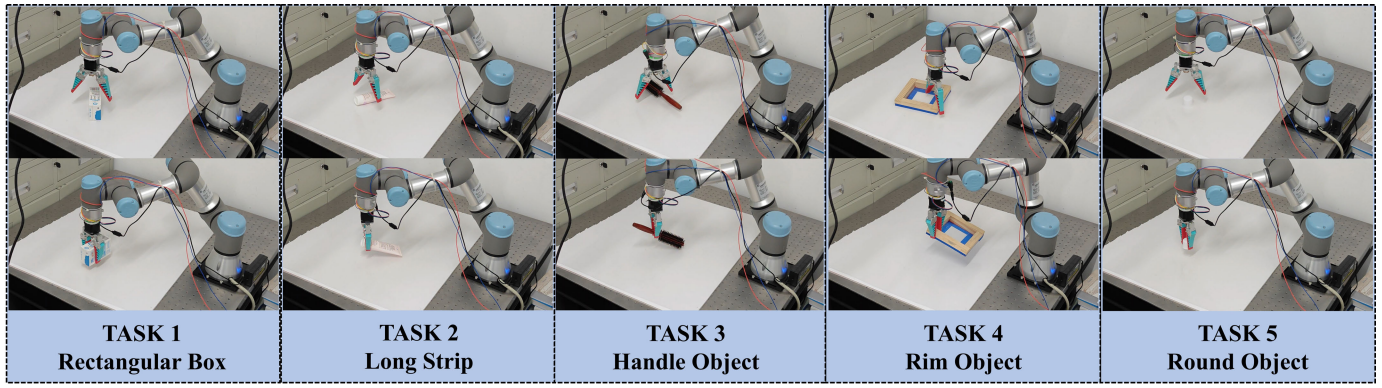


Fig. 8. Robot conducts grasp tests on samples for each task. All grasping actions are autonomously performed by the robot.

provide feasible grasp pose planning. Moreover, the relative relationships between the average grasp success rates of each method are consistent with those shown in Table V, indicating that experiments based on the dataset have certain guiding significance for real-world grasping. However, we observed that the performance in Task 5 was generally poor. We posit that such errors primarily stem from structural limitations of the gripper's end-effector. When performing planar grasping with a two-finger gripper, a narrower horizontal width of the end-effector increases the required positioning accuracy for spherical or cylindrical objects, thereby raising the likelihood of relative slippage and grasp failure during manipulation. Even when a grasp posture generated in the pixel coordinate system appears feasible, physical execution of the grasp may still fail due to inaccuracies such as hand-eye calibration errors, inherent robotic system errors, and mechanical constraints of the gripper structure. We intend to refine and incorporate these factors in future efforts and to further improve the annotation of grasping poses by integrating existing datasets with actual grasping observations.

VI. CONCLUSION

The primary focus of this work is to achieve efficient utilization of RGB-D information within a visual sensing system while effectively addressing continual learning tasks. This work introduces a three-branch RGB-D fusion module designed to capture complex spatial correlations and valuable features between RGB and depth modalities. The architecture, DF-ConvNet, demonstrates highly competitive performance on the Cornell and Jacquard benchmarks compared to recent state-of-the-art methods. Besides, we develop a composite loss-driven knowledge distillation framework for continual learning in grasp detection models, which simultaneously integrates pixel-level feature representations and high-level abstract experience from the teacher model, thereby enabling the student network to balance current-task adaptability with reduced forgetting of past knowledge. Ablation studies conducted under continual learning settings rigorously evaluated the efficacy of each proposed structural component. In comparative evaluations, our framework attained the highest mean grasp accuracy among benchmarked approaches. Finally, these models are deployed to a physical robotic platform for real-world grasping trials. Demonstrating robust

generalizability, it achieved an average grasp success rate of 82% when handling unseen objects across five distinct task scenarios. Comparative analysis confirms that our approach significantly outperforms existing state-of-the-art methods in practical applications.

On the other hand, our study exhibits several limitations that warrant discussion. First, the proposed continual learning framework maintains high grasp accuracy yet shows room for improvement in alleviating knowledge forgetting issues. Second, when deployed on physical robotic systems, the gripper exhibits limited robustness in handling irregularly shaped objects, particularly spherical items. It could be attributed to the hand-eye calibration errors and inherent robotic system errors in the grasping detection system when applied to real physical environments (deviations between 2-D planar grasping models and 3-D spatial grasping). It could also be related to the design of the end-effector's geometry. Third, within grasp detection, certain object categories present particular challenges, such as transparent and nonrigid items. Successful grasping of these objects depends critically on employing sensors extending beyond RGB principles and grippers with robust mechanical properties. Future work will focus on enhancing our system's performance for these specific cases through further experimentation and refinement. Fourth, while trained on established datasets like Cornell, our method requires further exploration to adapt to diverse continual learning scenarios involving dynamic or unpredictable environments. These constraints highlight opportunities for future refinement through enhanced knowledge retention mechanisms, gripper design optimizations, and dataset diversification strategies.

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